BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 062 · Programming Challenge and Review

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A £5-per-ticket program uses total = tickets + 5. What is the correction?

Q2. Insert 6 into sorted [2, 5, 8]. What is the result?
A. [2, 5, 8, 6]
B. [2, 5, 6, 8]
C. [6, 2, 5, 8]
D. [8, 6, 5, 2]

Q3. “A comparison operator must always produce a text paragraph.” Correct this. Explain why.

Q4. A warning needs high temperature OR high pressure. “OR is false when both inputs are true.” Correct this. What happens when both are high?

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I Do · Read new knowledge

Key vocabulary

Planning — Define inputs, processing, output and success criteria.

Selection — Use IF or ELSE to choose valid outcomes.

Iteration — Repeat actions with a bounded FOR or condition-controlled WHILE.

Testing — Check normal, boundary and invalid inputs, then refine.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

Planning: Define inputs, processing, output and success criteria. For example, consider a club booking tool.

Selection: Use IF or ELSE to choose valid outcomes. For example, consider a revision calculator.

Iteration: Repeat actions with a bounded FOR or condition-controlled WHILE. For example, consider a rewards app.

Testing: Check normal, boundary and invalid inputs, then refine. For example, consider a canteen voucher.

Worked example

A club booking tool is being developed. Explain how planning and selection are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Planning [1]; Define inputs, processing, output and success criteria. [1].

3. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A revision calculator is being programmed. Explain why planning matters. [2 marks]

Worked method: Planning [1]; Define inputs, processing, output and success criteria. [1].

Guided 2

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For a rewards app, explain how selection is used. [2 marks]

Worked method: Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

Guided 3

A canteen voucher needs iteration and testing. Explain the roles of both. [4 marks]

Worked method: Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing condition to report a hot sensor.

```
sensor = 29
if ... then
    print("Too hot")
else
    print("OK")
endif
```

write

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how selection and iteration can help a fundraising tickets. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of planning, iteration and testing when creating a quiz score. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

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Exit Ticket · Answer before checking

A useful algorithm plan identifies inputs, processing, outputs and success criteria.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Use total = tickets * 5; addition does not calculate £5 for each ticket.

Q2. B — [2, 5, 6, 8]

Q3. A Boolean result: true or false.

Q4. OR is true if at least one input is true, including both. AND needs both true. NOT reverses a value.

I Do / We Do pages · Guided practice

Q1. Planning [1]; Define inputs, processing, output and success criteria. [1].

Q2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

Q3. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

You Do pages · Code First

```
sensor = 29
if sensor > 28 then
    print("Too hot")
else
    print("OK")
endif
```

Expected result: Too hot

You Do pages · Independent questions

Q1. Planning [1]; Iteration [1].

Q2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1].

Q3. Planning [1]; Define inputs, processing, output and success criteria. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: